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BSc in Applied Data Science Communication

Comparative Corpus Analysis

Topic: The Tea Industry

Genre Comparison: Research Articles vs. Reddit Comments

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LB 3144: Corpus Analysis & Data Presentation

Year 3 : Semester 5

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1. INTRODUCTION

In this report, the corpus analysis of the tea industry is conducted in two different genres, namely, scholarly/research articles and comments on Reddit. The academic and industry magazines and newspapers on tea production, trade, and economics were consulted as sources of research articles (10 texts - 67,000 tokens). The collection of Reddit comments was based on subreddits dedicated to teas with consumer-related discussions (10 texts- approximately 2,000 tokens). Tokens are the single word occurrences. The frequency and dispersion analysis was achieved in LancsBox 5.5.1 with the help of a frequency and dispersion tool called Whelk, a contextual patterns tool called KWIC, and a collocation networks tool called GraphColl.

2. FREQUENCY ANALYSIS

2.1 Most Frequency keyword per genre

The table below demonstrates the best identified keywords by LancsBox per corpus.

Research Articles keywords	Reddit comments keywords
tea	tea
industry	Ceylon
production	taste
Export	flavor
Market	brand
Innovation	price
Plantation	drink
Value-added	quality
Processing	brew
customer	loose leaf

Table 1: Keyword Comparison

The industry-level terms dominate in research articles: ***industry, production, export, market, innovation, plantation*** and ***customer***. These are indicators of an economic and policy driven packaging of tea as a commodity across the world. Conversely, the key words used on the Reddit site, ***Ceylon, taste, flavor, brand, price, brew, quality, loose leaf***, show that the discussion is consumer-oriented, which is concentrated on a sensory experience and a buying choice.

Comparative Corpus Analysis - Research Articles vs. Reddit

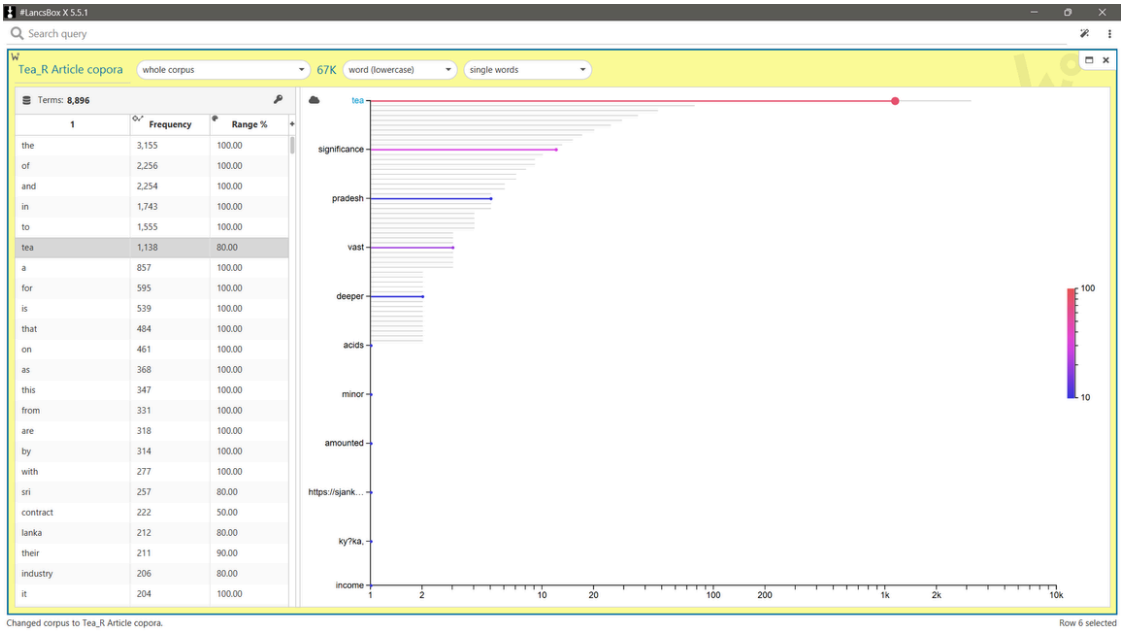


Figure1: Research Articles - Word Frequency List & Dispersion Chart

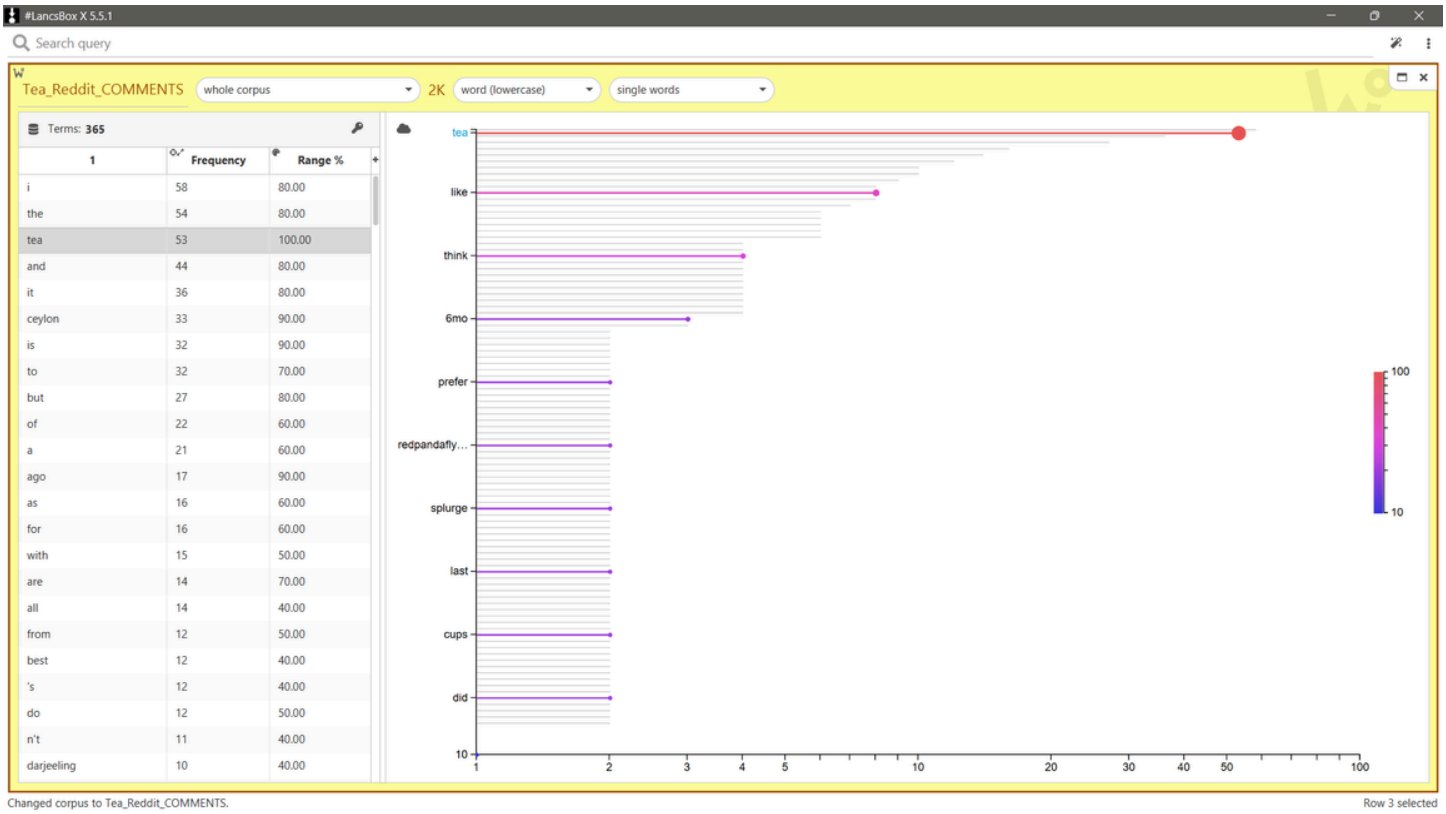


Figure 2: Reddit Comments - Word Frequency List & Dispersion Chart

2.2 Word Frequency Lists

Figure 1 and Figure 2 show the complete list of frequencies and dispersion bar charts of the Whelk tool of each corpus.

Tea is 7th in the corpus of articles (67K tokens, 8,896 unique terms) with a frequency of 1,138 and a Range of 80%, means 8/10 articles it appears. The top is dominated by use of structural words (the, of, and), which are also typical of formal academic writing. On the Reddit corpus (2K tokens, 365 unique terms), tea is positioned in the 3rd position with frequency 53 and the optimal Range of 100% as it is mentioned in all of the posts, the text on Reddit is also directly centered around the topic.

2.3 Side by side Frequency Comparison

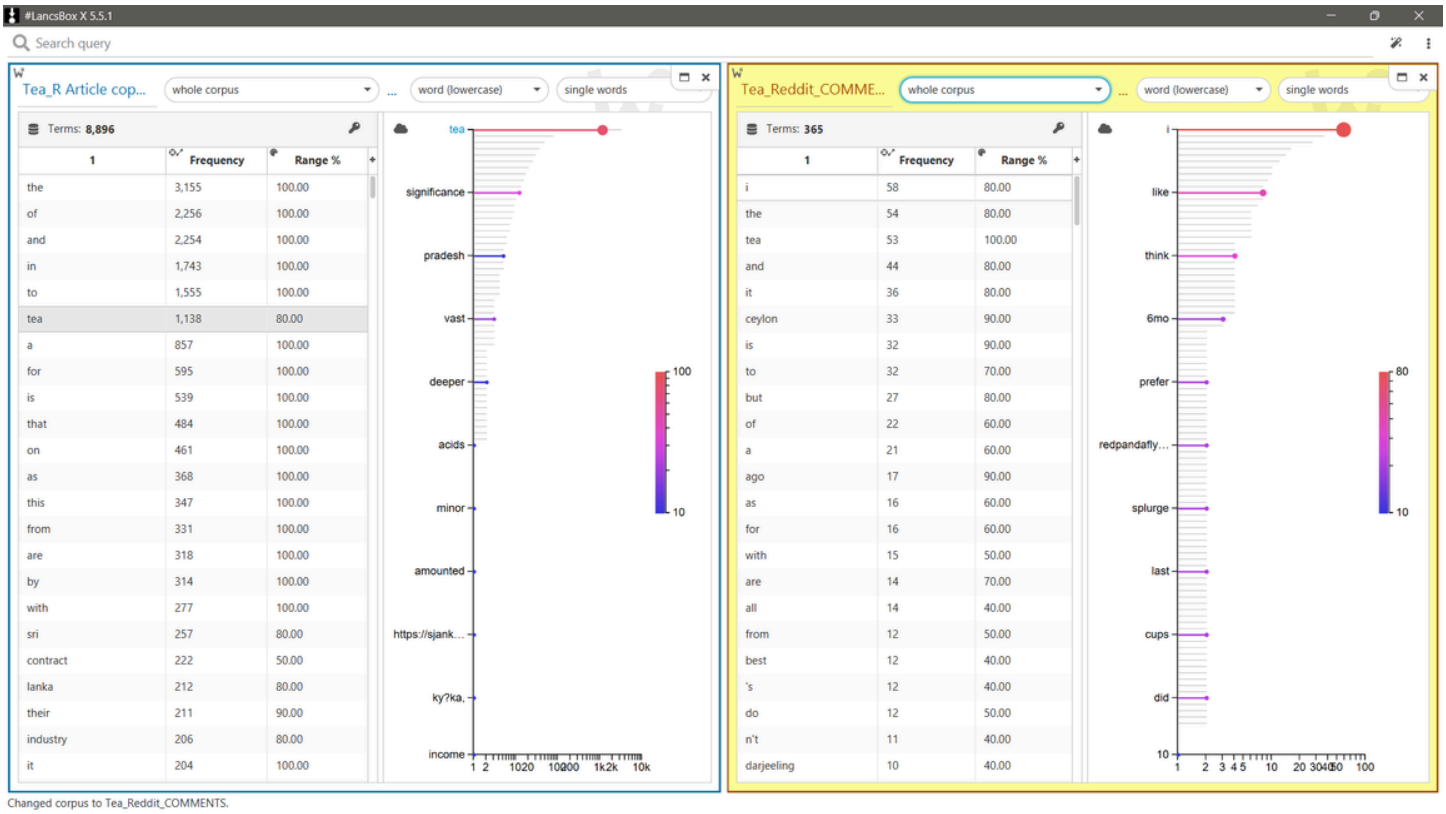


Figure 3: Both corpora displayed side-by-side in LancsBox

3. DISPERSION ANALYSIS

3.1 Word Clouds

Figures 4 and 5 are word cloud plots at LancsBox in which word size represents frequency and the colour of the word represents Range % (blue = low, red = high).

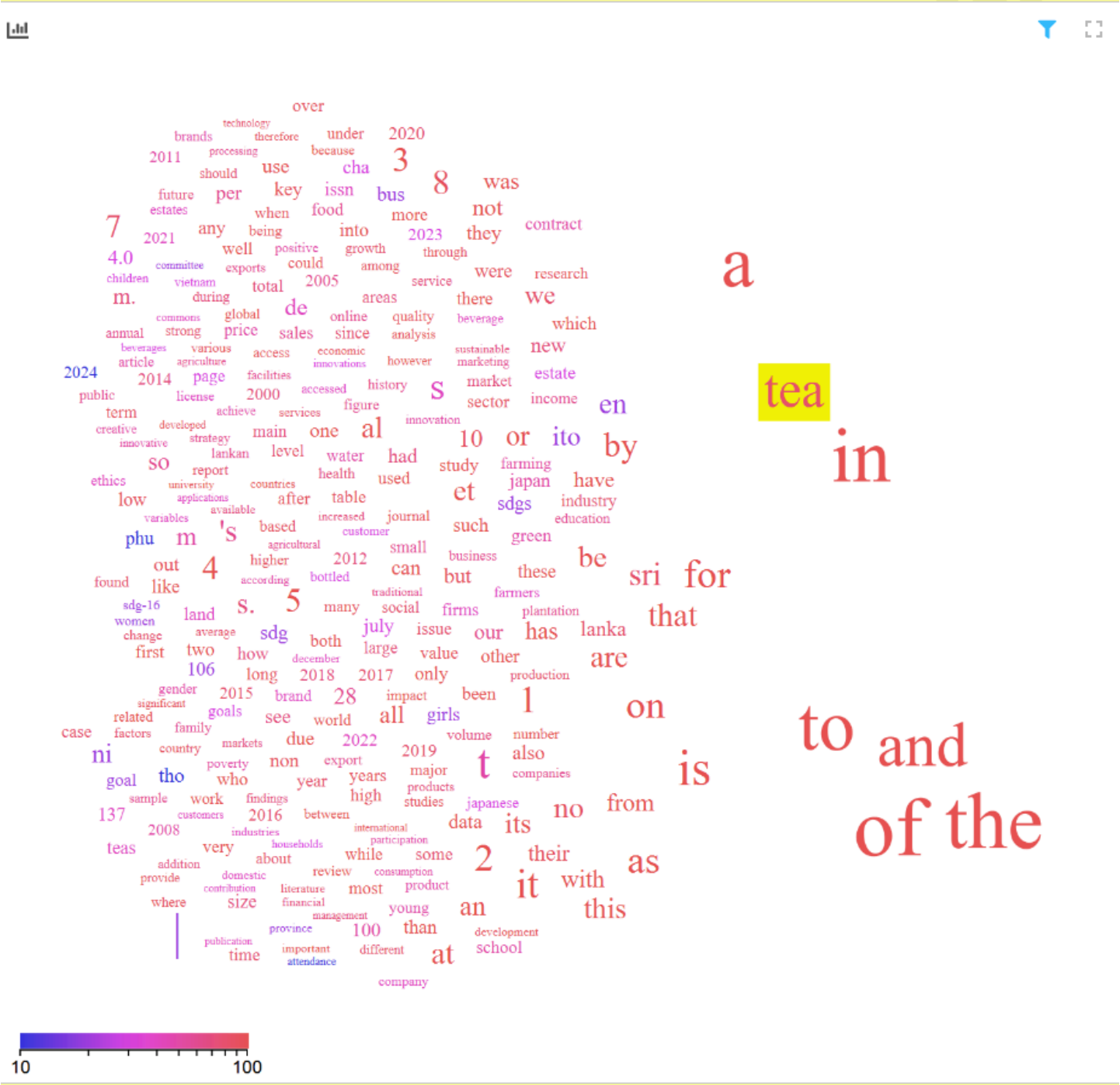


Figure 4: Research Articles Word Cloud

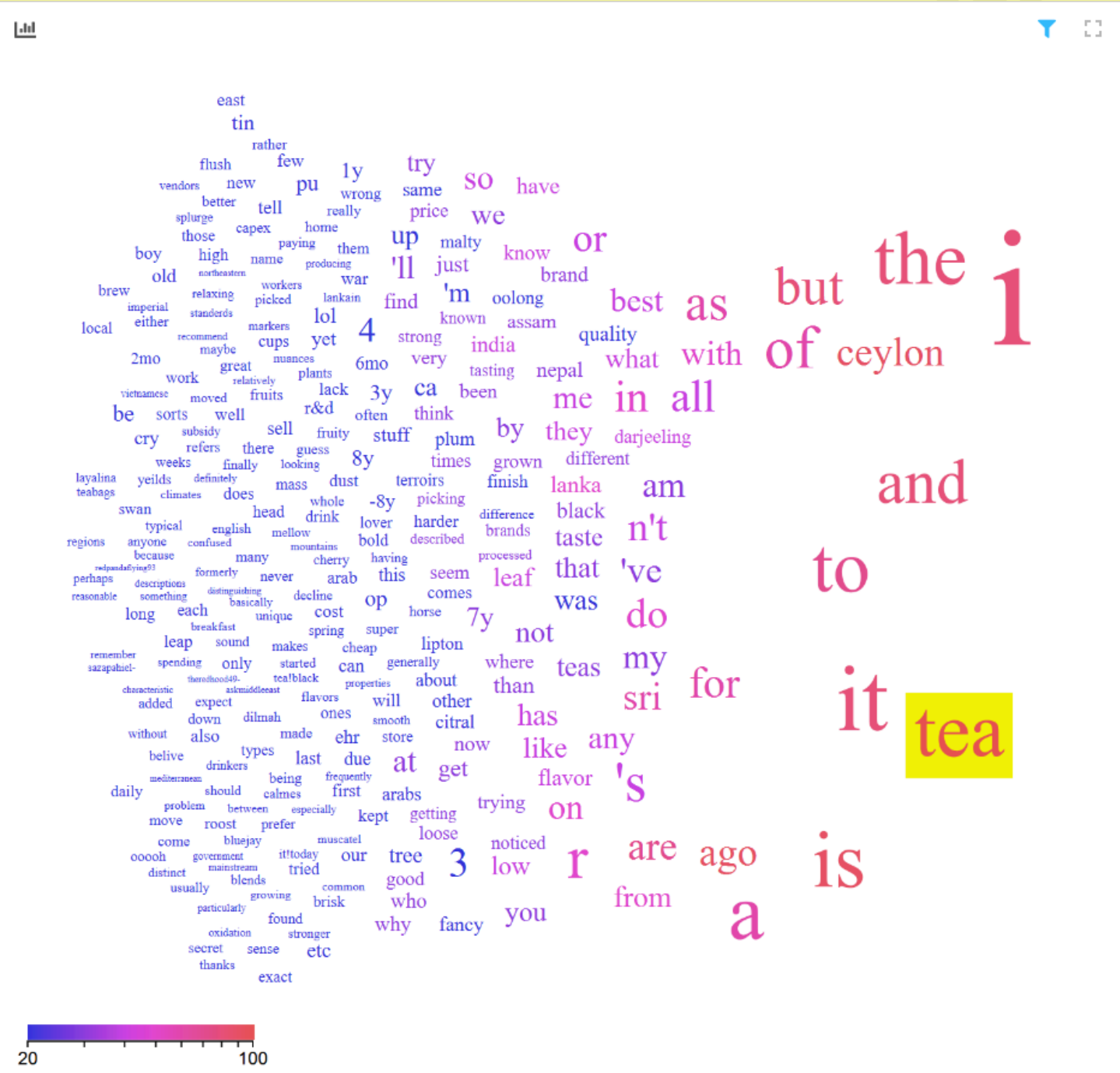


Figure 5: Reddit Comments Word Cloud

Figure 4 (articles) shows that tea is the most predominant word (red, greatest dispersion), in the middle of pink/purple mid-range words, there is industry terms such as ***sri, lanka, production, industry.*** The vocabulary of the word cloud is rich and diverse, which indicates the range of academic words. The cloud in Figure 5 (Reddit) is smaller and is more concentrated on consumer terms: ***ceylon, taste, flavor, brew, quality, brand***. The personal and first-person character of the Reddit discourse is proven by the word i, which is put in red (100% Range, used in all posts).

4. THE KWIC AND COLLCATION ANALYSIS

LancsBox KWIC and GraphColl tools were used to analyze six terms - three terms each corpus. KWIC displays every word in the context of other words; GraphColl plots most common collocates.

4.1 Industry, Export and Production Research Articles.

Industry - Freq.: 206 (3,090.17), Texts: 8/10

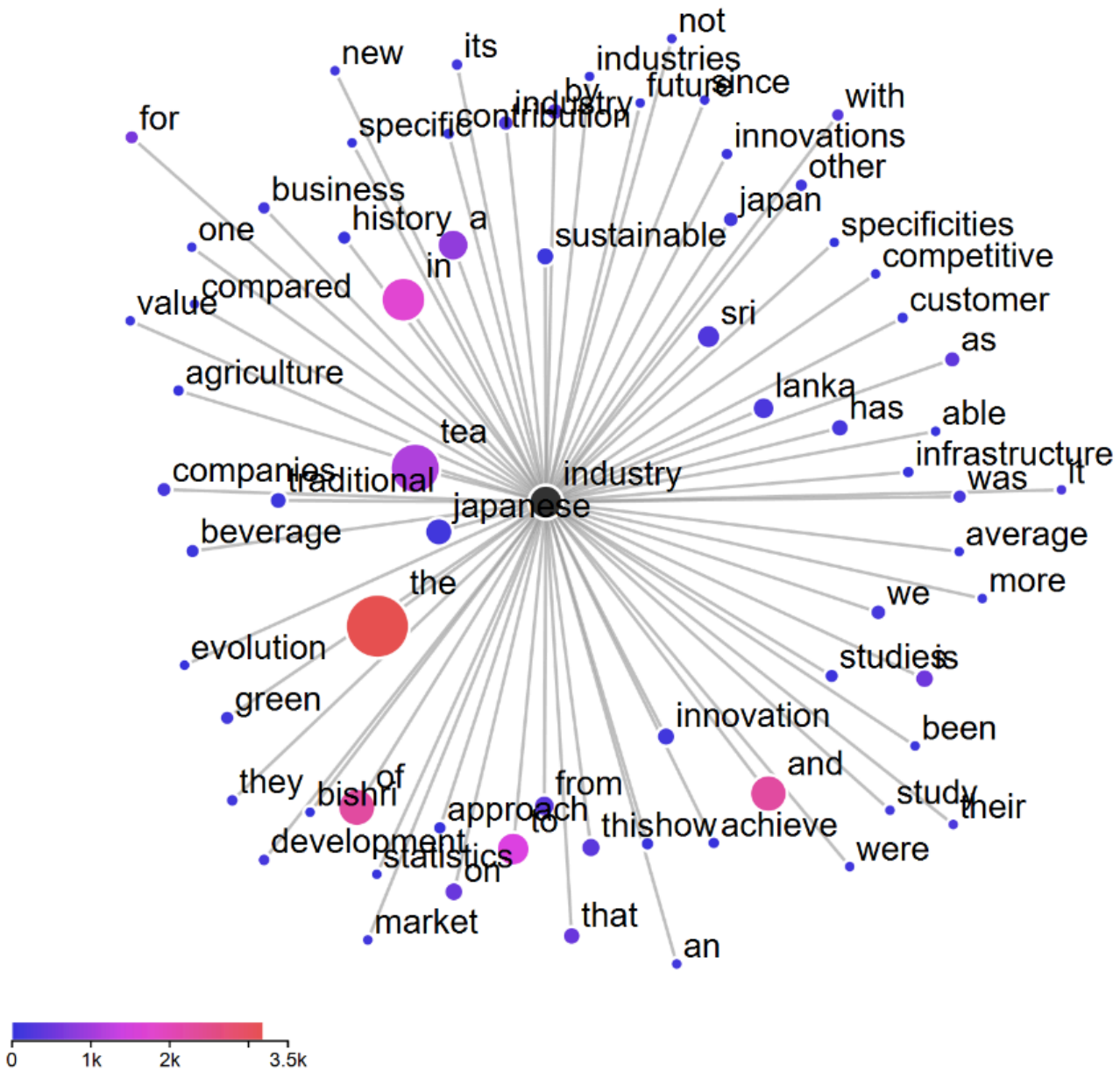
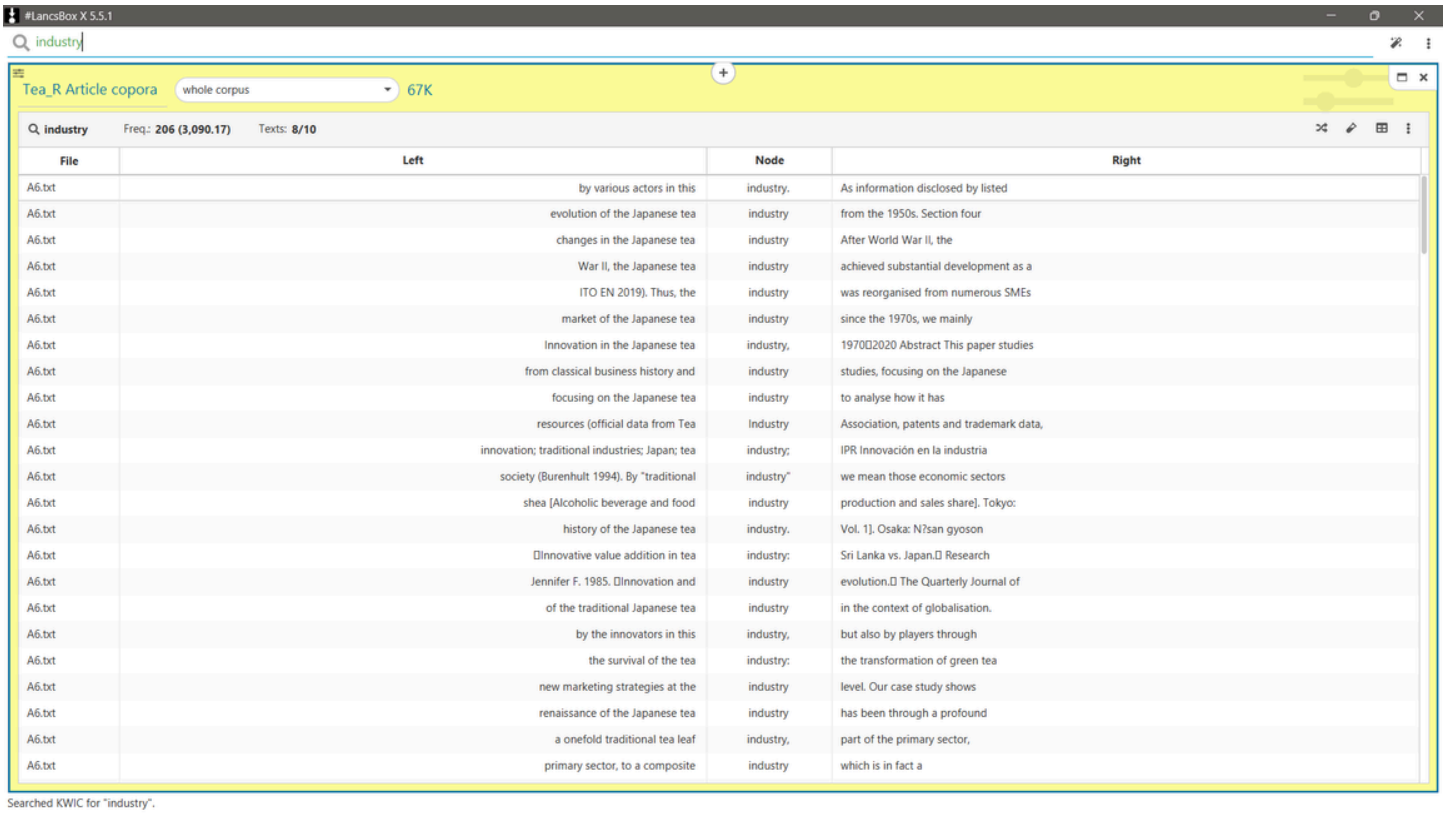


Figure 6: KWIC results for "industry" in Research Articles



File	Left	Node	Right
A6.txt	by various actors in this	industry.	As information disclosed by listed
A6.txt	evolution of the Japanese tea	industry	from the 1950s. Section four
A6.txt	changes in the Japanese tea	industry	After World War II, the
A6.txt	War II, the Japanese tea	industry	achieved substantial development as a
A6.txt	ITO EN 2019). Thus, the	industry	was reorganised from numerous SMEs
A6.txt	market of the Japanese tea	industry	since the 1970s, we mainly
A6.txt	Innovation in the Japanese tea	industry,	1970□2020 Abstract This paper studies
A6.txt	from classical business history and	industry	studies, focusing on the Japanese
A6.txt	focusing on the Japanese tea	industry	to analyse how it has
A6.txt	resources (official data from Tea	Industry	Association, patents and trademark data,
A6.txt	innovation; traditional industries; Japan; tea	industry;	IPR Innovación en la industria
A6.txt	society (Burenhult 1994). By "traditional	industry"	we mean those economic sectors
A6.txt	shea [Alcoholic beverage and food	industry	production and sales share]. Tokyo:
A6.txt	history of the Japanese tea	industry.	Vol. 1]. Osaka: N?san gyoson
A6.txt	□Innovative value addition in tea	industry:	Sri Lanka vs. Japan.□ Research
A6.txt	Jennifer F. 1985. □Innovation and	industry	evolution.□ The Quarterly Journal of
A6.txt	of the traditional Japanese tea	industry	in the context of globalisation.
A6.txt	by the innovators in this	industry,	but also by players through
A6.txt	the survival of the tea	industry:	the transformation of green tea
A6.txt	new marketing strategies at the	industry	level. Our case study shows
A6.txt	renaissance of the Japanese tea	industry	has been through a profound
A6.txt	a onefold traditional tea leaf	industry,	part of the primary sector,
A6.txt	primary sector, to a composite	industry	which is in fact a

Figure 7: GraphColl collocation network - industry

The industry is mentioned 206 times in 8 of 10 articles (Range 80%) which validates it being a central and frequently reoccurring theme. It is seen in KWIC lines with tea always, in the institutional occurrence: "evolution of the Japanese tea industry", "Innovation in the Japanese tea industry". The high density of collocation links to japanese, tea, traditional, business, sustainable, innovation, market, development - representing an academic situationalization of tea as an economic and historical phenomenon.

Export - Freq.: 51 (765.04), Texts: 6/10

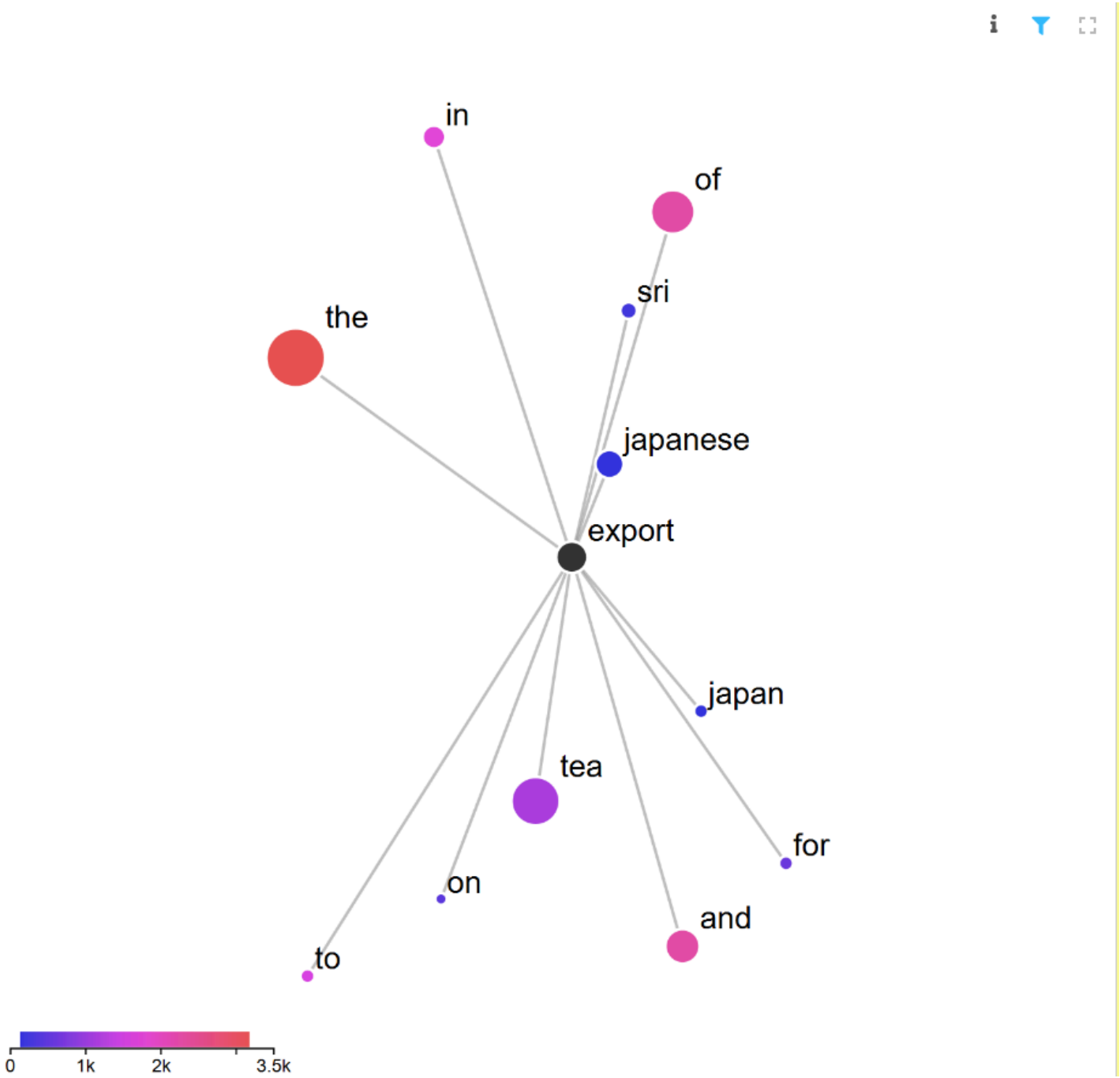
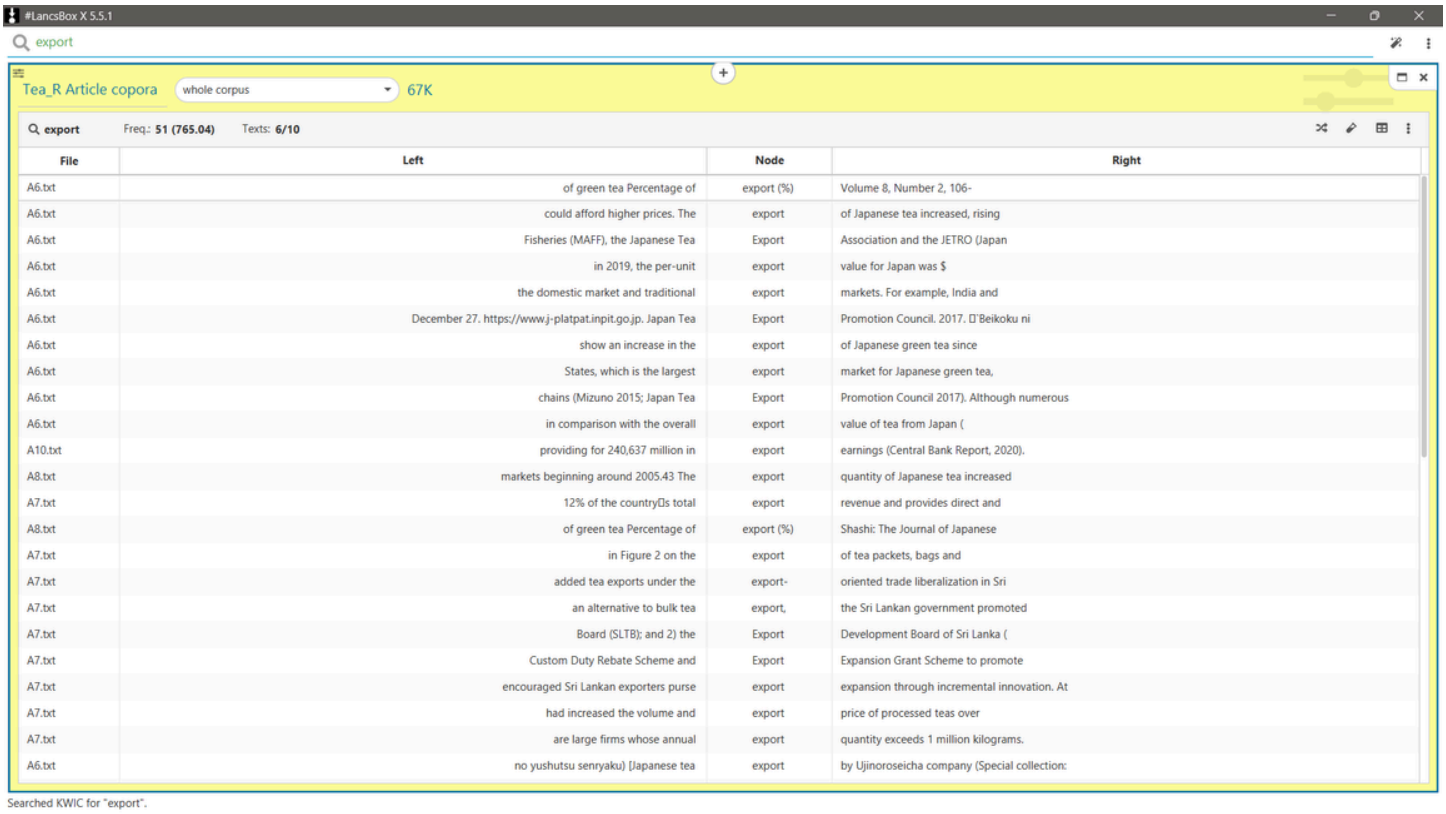


Figure 8: KWIC results for "export" in Research Articles corpus



File	Left	Node	Right
A6.txt	of green tea Percentage of	export (%)	Volume 8, Number 2, 106-
A6.txt	could afford higher prices. The	export	of Japanese tea increased, rising
A6.txt	Fisheries (MAFF), the Japanese Tea	Export	Association and the JETRO (Japan
A6.txt	in 2019, the per-unit	export	value for Japan was $
A6.txt	the domestic market and traditional	export	markets. For example, India and
A6.txt	December 27. https://www.j-platpat.inpit.go.jp. Japan Tea	Export	Promotion Council. 2017. □'Beikoku ni
A6.txt	show an increase in the	export	of Japanese green tea since
A6.txt	States, which is the largest	export	market for Japanese green tea,
A6.txt	chains (Mizuno 2015; Japan Tea	Export	Promotion Council 2017). Although numerous
A6.txt	in comparison with the overall	export	value of tea from Japan (
A10.txt	providing for 240,637 million in	export	earnings (Central Bank Report, 2020).
A8.txt	markets beginning around 2005.43 The	export	quantity of Japanese tea increased
A7.txt	12% of the country□s total	export	revenue and provides direct and
A8.txt	of green tea Percentage of	export (%)	Shashi: The Journal of Japanese
A7.txt	in Figure 2 on the	export	of tea packets, bags and
A7.txt	added tea exports under the	export-	oriented trade liberalization in Sri
A7.txt	an alternative to bulk tea	export,	the Sri Lankan government promoted
A7.txt	Board (SLTB); and 2) the	Export	Development Board of Sri Lanka (
A7.txt	Custom Duty Rebate Scheme and	Export	Expansion Grant Scheme to promote
A7.txt	encouraged Sri Lankan exporters purse	export	expansion through incremental innovation. At
A7.txt	had increased the volume and	export	price of processed teas over
A7.txt	are large firms whose annual	export	quantity exceeds 1 million kilograms.
A6.txt	no yushutsu senryaku) [Japanese tea	export	by Ujinoroseicha company (Special collection:

Figure 9: GraphColl collocation network for "export"

Export is 51 times and the lower Range of 60, which is concentrated in trade oriented articles. KWIC validates the economic and policy backgrounds: export of Japanese tea increased, Export Development Board of Sri Lanka. The collocation graph relates the most to the ***japanese, sri, tea, japan*** making the discourse of exportation rely on national tea production.

Production - Freq.: 137 (2,055.11), Texts: 8/10

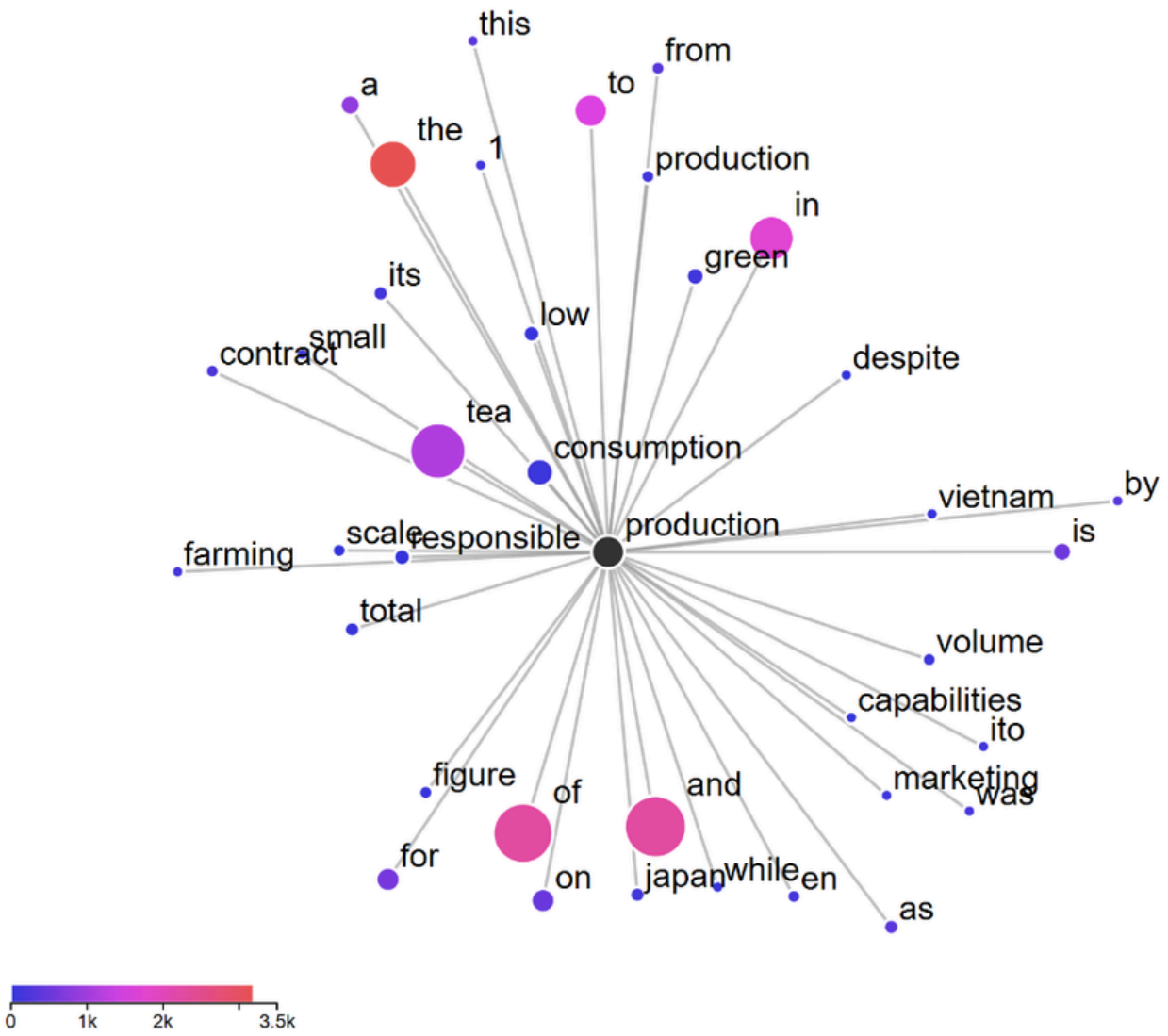
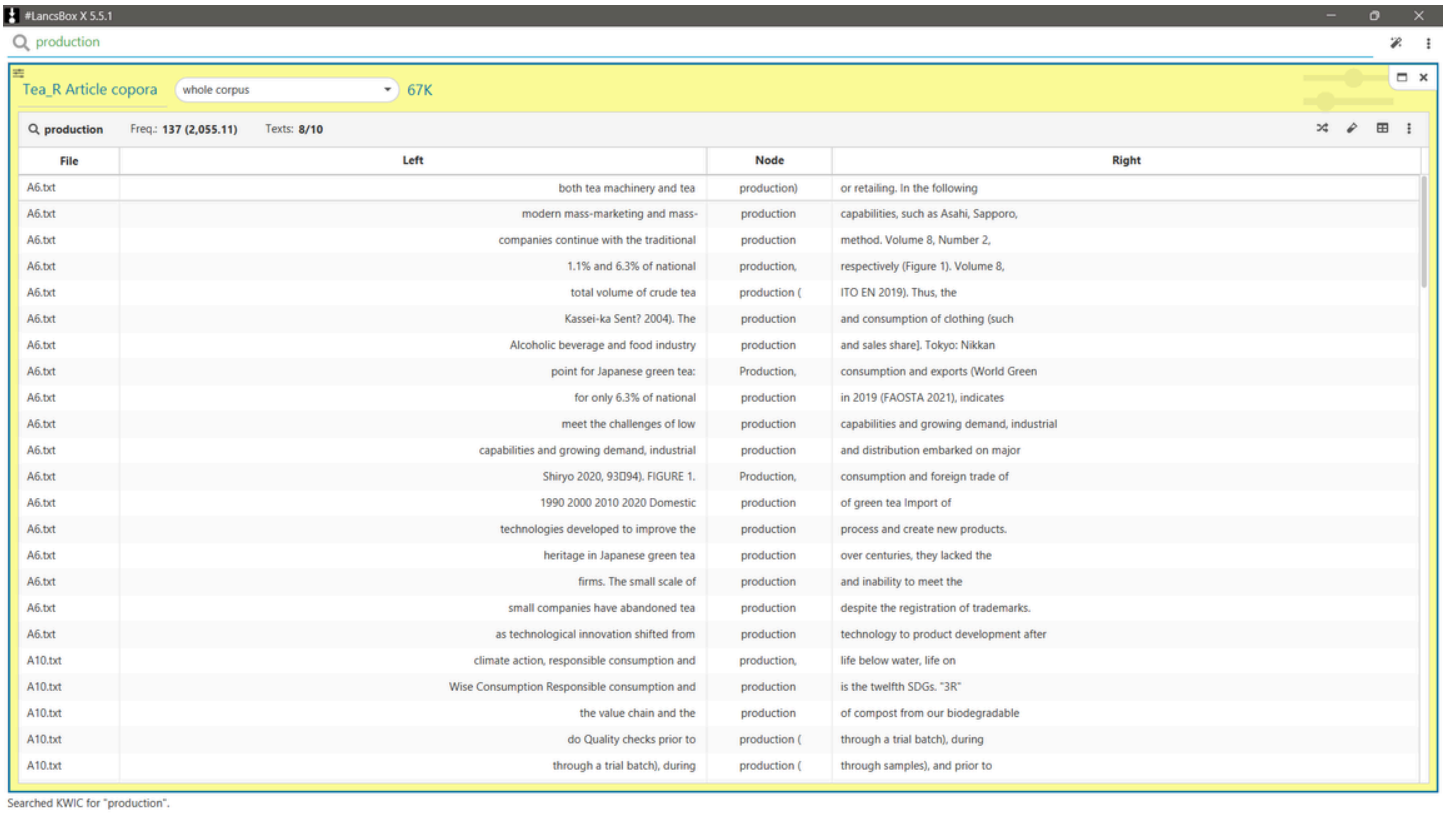


Figure 10: KWIC results for "production" in Research Articles corpus



File	Left	Node	Right
A6.txt	both tea machinery and tea	production)	or retailing. In the following
A6.txt	modern mass-marketing and mass-	production	capabilities, such as Asahi, Sapporo,
A6.txt	companies continue with the traditional	production	method. Volume 8, Number 2,
A6.txt	1.1% and 6.3% of national	production,	respectively (Figure 1). Volume 8,
A6.txt	total volume of crude tea	production (	ITO EN 2019). Thus, the
A6.txt	Kassei-ka Sent? 2004). The	production	and consumption of clothing (such
A6.txt	Alcoholic beverage and food industry	production	and sales share]. Tokyo: Nikkan
A6.txt	point for Japanese green tea:	Production,	consumption and exports (World Green
A6.txt	for only 6.3% of national	production	in 2019 (FAOSTA 2021), indicates
A6.txt	meet the challenges of low	production	capabilities and growing demand, industrial
A6.txt	capabilities and growing demand, industrial	production	and distribution embarked on major
A6.txt	Shiryo 2020, 93□94). FIGURE 1.	Production,	consumption and foreign trade of
A6.txt	1990 2000 2010 2020 Domestic	production	of green tea Import of
A6.txt	technologies developed to improve the	production	process and create new products.
A6.txt	heritage in Japanese green tea	production	over centuries, they lacked the
A6.txt	firms. The small scale of	production	and inability to meet the
A6.txt	small companies have abandoned tea	production	despite the registration of trademarks.
A6.txt	as technological innovation shifted from	production	technology to product development after
A10.txt	climate action, responsible consumption and	production,	life below water, life on
A10.txt	Wise Consumption Responsible consumption and	production	is the twelfth SDGs. "3R"
A10.txt	the value chain and the	production	of compost from our biodegradable
A10.txt	do Quality checks prior to	production (	through a trial batch), during
A10.txt	through a trial batch), during	production (	through samples), and prior to

Figure 11: GraphColl collocation network for "production"

There are also 8 articles that list production 137 times (Range 80%). KWIC unveils the supply-chain framing: "the amount of crude tea that has been produced in total volume, technologies that have been created to enhance the production process. Its collocates include consumption, green, farming, responsible, volume, vietnam, japan - tracking the entire agricultural supply line. It is worth noting that there is no ‘production’ in the world of Reddit, which highlights the producer-vs-consumer gap between genres.

4.2 Reddit Comments: Ceylon, Taste & Brand

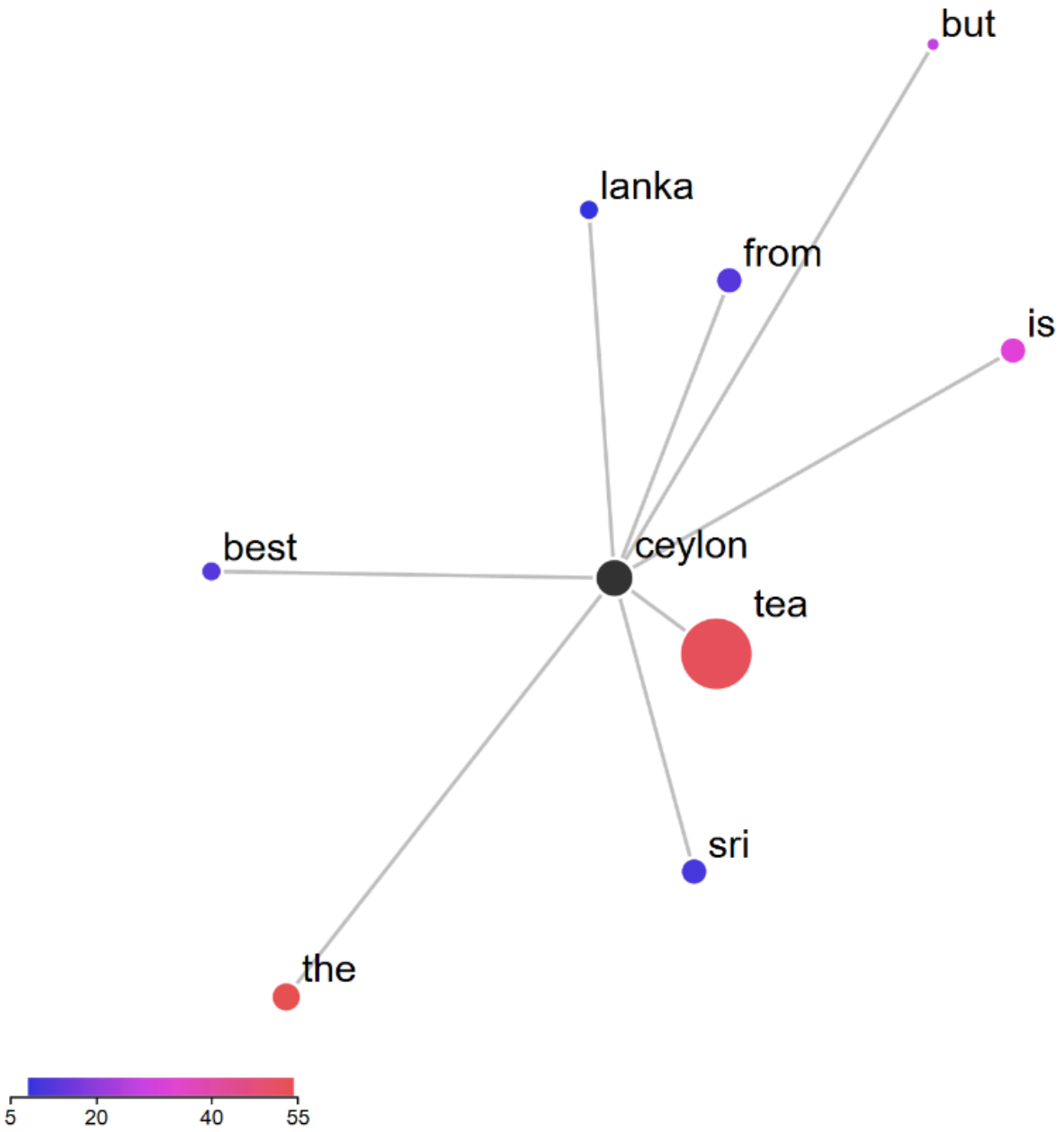
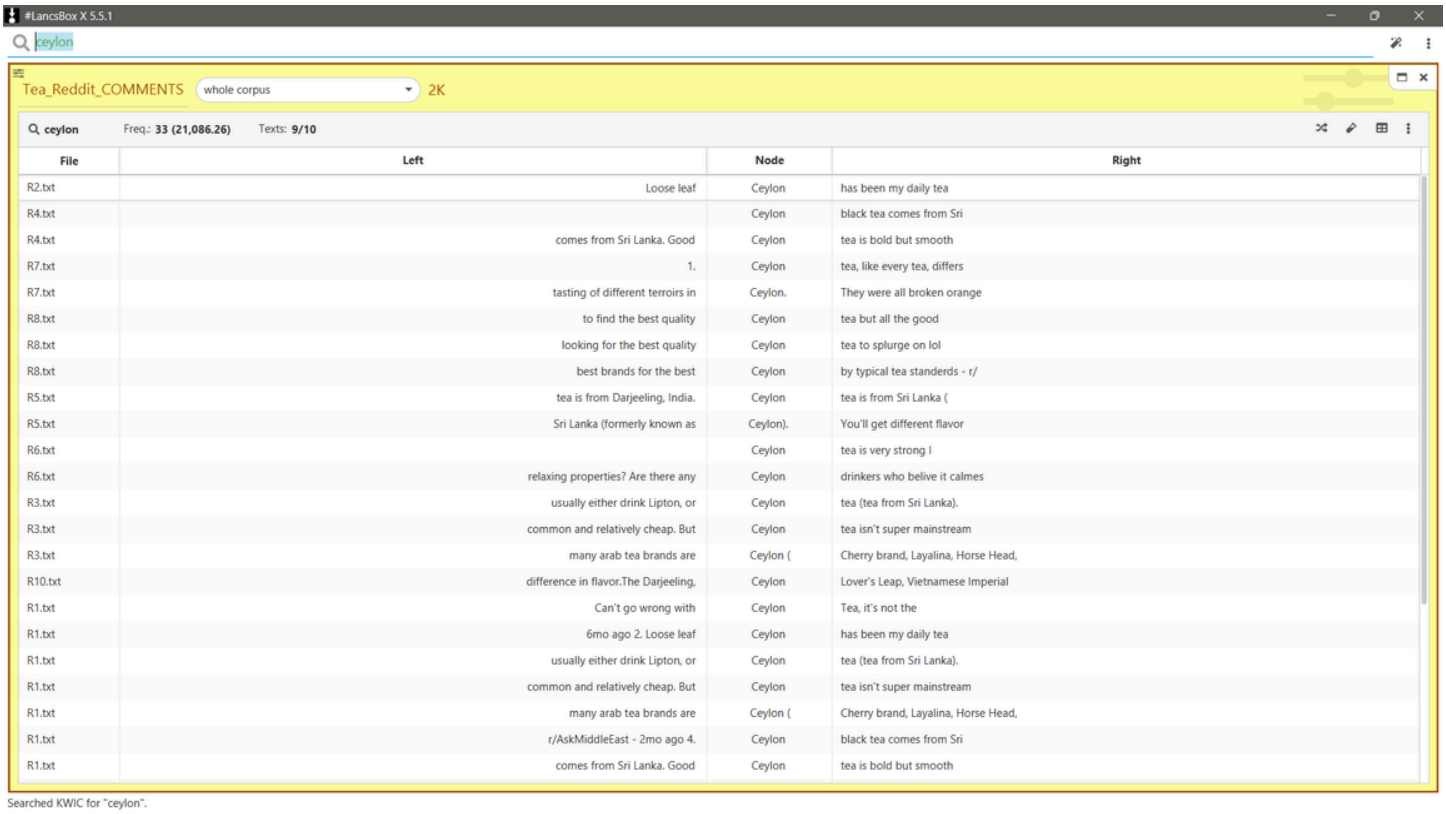


Figure 12: KWIC results for "ceylon" in Reddit Comments corpus



File	Left	Node	Right
R2.txt	Loose leaf	Ceylon	has been my daily tea
R4.txt		Ceylon	black tea comes from Sri
R4.txt	comes from Sri Lanka. Good	Ceylon	tea is bold but smooth
R7.txt	1.	Ceylon	tea, like every tea, differs
R7.txt	tasting of different terroirs in	Ceylon.	They were all broken orange
R8.txt	to find the best quality	Ceylon	tea but all the good
R8.txt	looking for the best quality	Ceylon	tea to splurge on lol
R8.txt	best brands for the best	Ceylon	by typical tea standerds - r/
R5.txt	tea is from Darjeeling, India.	Ceylon	tea is from Sri Lanka (
R5.txt	Sri Lanka (formerly known as	Ceylon).	You'll get different flavor
R6.txt		Ceylon	tea is very strong I
R6.txt	relaxing properties? Are there any	Ceylon	drinkers who belive it calmes
R3.txt	usually either drink Lipton, or	Ceylon	tea (tea from Sri Lanka).
R3.txt	common and relatively cheap. But	Ceylon	tea isn't super mainstream
R3.txt	many arab tea brands are	Ceylon (	Cherry brand, Layalina, Horse Head,
R10.txt	difference in flavor.The Darjeeling,	Ceylon	Lover's Leap, Vietnamese Imperial
R1.txt	Can't go wrong with	Ceylon	Tea, it's not the
R1.txt	6mo ago 2. Loose leaf	Ceylon	has been my daily tea
R1.txt	usually either drink Lipton, or	Ceylon	tea (tea from Sri Lanka).
R1.txt	common and relatively cheap. But	Ceylon	tea isn't super mainstream
R1.txt	many arab tea brands are	Ceylon (	Cherry brand, Layalina, Horse Head,
R1.txt	r/AskMiddleEast - 2mo ago 4.	Ceylon	black tea comes from Sri
R1.txt	comes from Sri Lanka. Good	Ceylon	tea is bold but smooth

Figure 13: GraphColl collocation network for "ceylon"

The KWIC lines demonstrate that ceylon is always applied in the consumer environments: "Loose leaf Ceylon has been my tea daily, Ceylon tea is smack-Down but mellow, best quality Ceylon tea. It acts as a marker of quality of origin. The collocation network makes it clear that its nearest associates are ***tea, sri, lanka, from, best***, and all of them are related to origin and quality assessment. This is very different with the articles where we find the term Sri Lanka in the economic and trade set-ups.

4.3 Taste (Reddit)

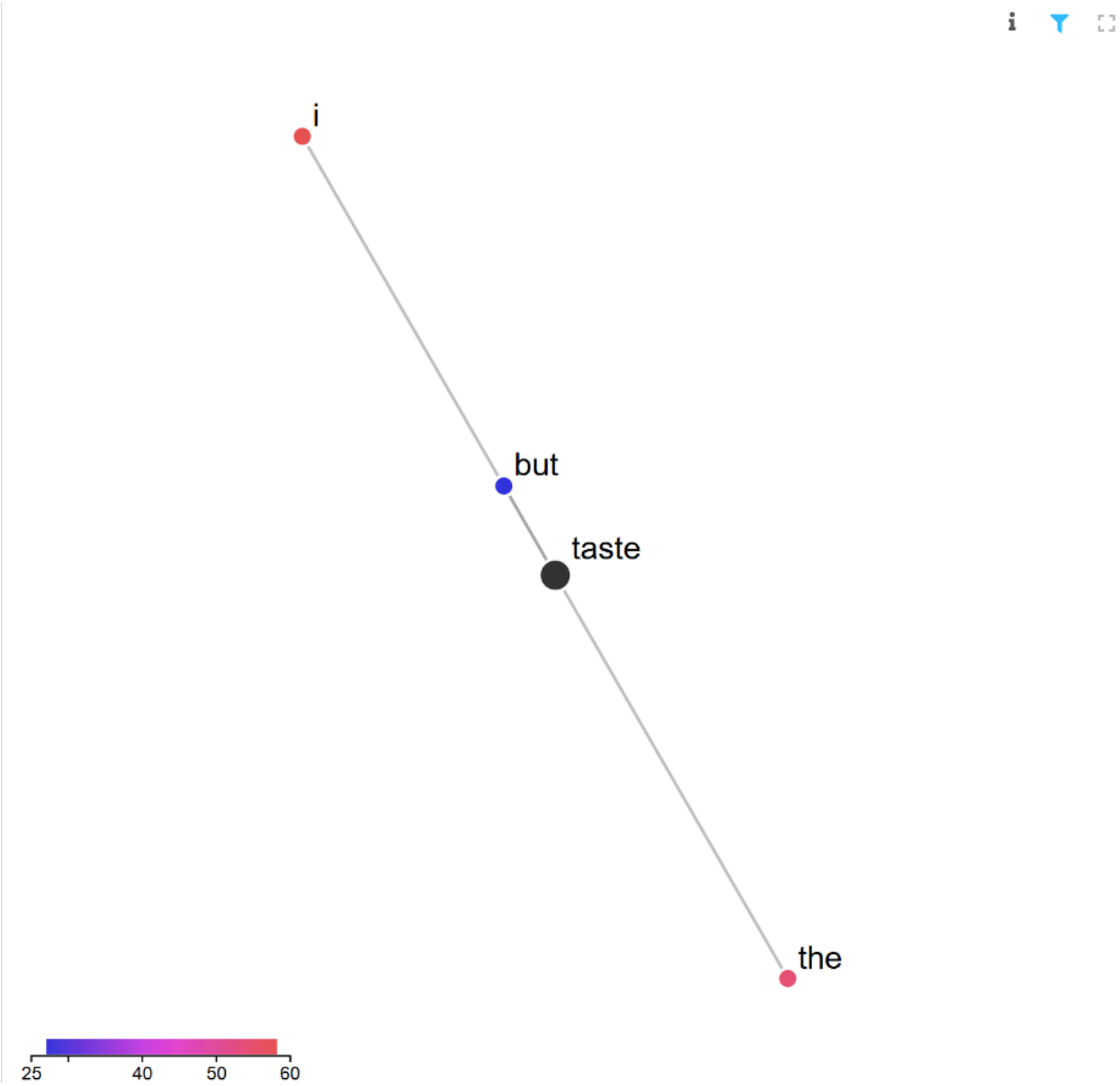
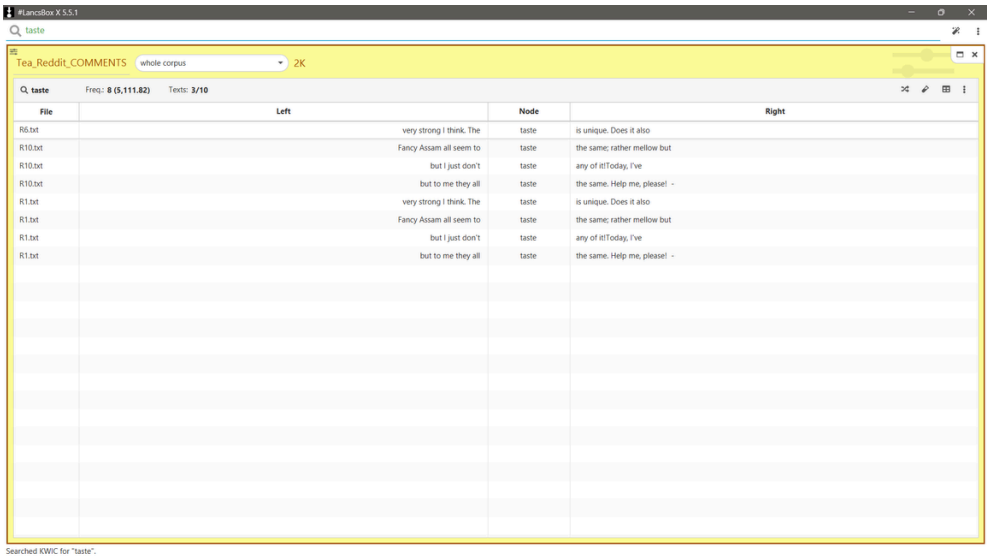


Figure 14: KWIC results for "taste" in Reddit Comments corpus



#LancsBox X 5.5.1

taste

Tea_Reddit_COMMENTS | whole corpus | 2K

taste | Freq.: 8 (5,111.82) | Texts: 3/10

File	Left	Node	Right
R6.txt	very strong I think. The	taste	is unique. Does it also
R10.txt	Fancy Assam all seem to	taste	the same; rather mellow but
R10.txt	but I just don't	taste	any of it!Today, I've
R10.txt	but to me they all	taste	the same. Help me, please! -
R1.txt	very strong I think. The	taste	is unique. Does it also
R1.txt	Fancy Assam all seem to	taste	the same; rather mellow but
R1.txt	but I just don't	taste	any of it!Today, I've
R1.txt	but to me they all	taste	the same. Help me, please! -

Searched KWIC for "taste".

Figure 15: GraphColl collocation network for "taste"

Taste is found 8 times (Freq. 5,111.82) in 3/10 Reddit texts (Range 30%) only, so there is low dispersion - Taste occurs particularly in posts that are specifically about the appraisal of teas. It is indicated in KWIC line in subjective reviews: The taste is unique, they all are the same. The collocation graph represents a insignificant number of ties (only i, but, the) in line with brevity and individual preference taste-description sentences.

4.5 Brand (Reddit)

Figure 16: KWIC results for "brand"

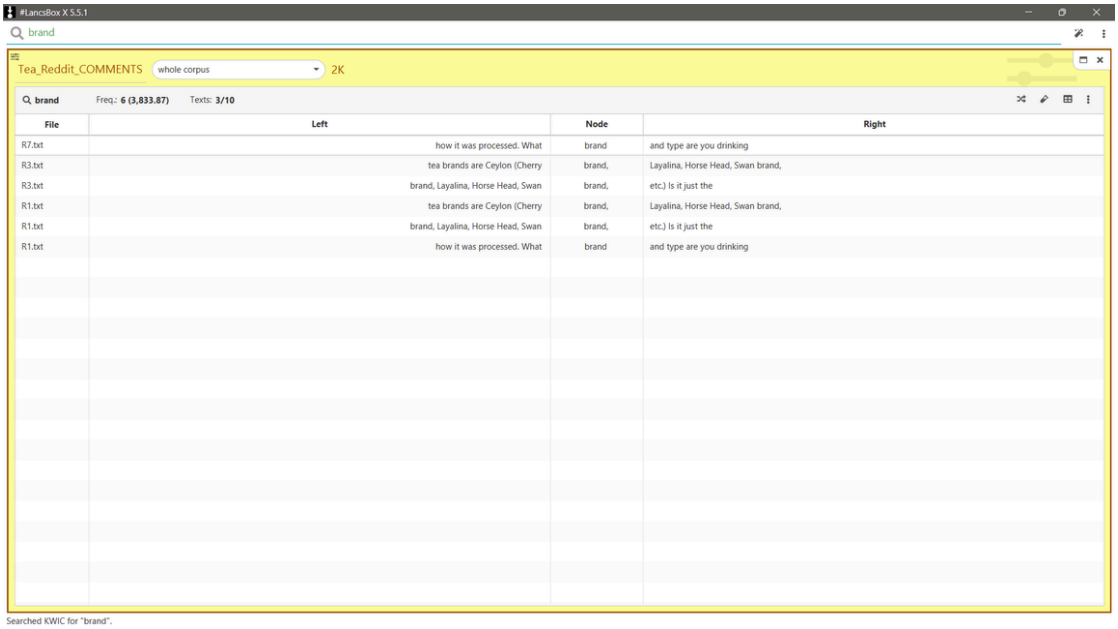


Figure 17: GraphColl collocation network for "brand"

Brand appears 6 (Freq. 3,833.87) times in 3 (Range 30%) texts. It appears in product-comparison contexts in KWIC lines: "tea brands are Ceylon (Cherry brand, Layalina, Horse Head, Swan brand), this implies that consumers are in fact engaging in comparison of the commercial tea brands. Its collocation network is related to ***ceylon, taste, Sri Lanka, best*** showing that brand discussion is mixed with its origin and quality evaluation in the context of Reddit.

5. CONCLUSION

This discussion demonstrates that the lexical choice is influenced by genre. The central terms in both corpora are tea though their framework differs. The economic vocabulary in research articles takes the form of industry, export, production thus displays a supply-chain, academic view. At higher normalized frequencies but with reduced dispersion, Reddit comment foregrounds on consumer terms, taste, flavor, brand, ceylon, etc., but only in particular posts. KWIC analysis supports this, where "Ceylon" is a pointer of quality and provenience to the users of the Reddit, whereas Sri Lanka is a pointer of geography and commerce in scholarly literature. These disparities are a manifestation of the specific communicative goal of each genre: researchers are talking to policymakers and scholars, users of the Reddit community share their personal experience as a part of a community of consumers.

_____End of report_____